

II. **Apache Samza:** This is an open source distributed stream processing framework. It uses Apache Kafka for messaging and Apache Hadoop YARN to provide fault tolerance, security and resource management. Samza is built to handle a large amount of states.

III. **Apache Spark:** This open source platform is for large scale data processing. For cluster management, Spark supports Hadoop YARN or Apache Mesos. For distributed storage, it can use Hadoop Distributed File System (HDFS), MapR file system, Cassandra, Kudu etc. It also supports a pseudo-distributed local mode.

IV. **Apache Storm:** This distributed real-time computing system is fast, and easy to set up and operate. It is scalable and fault-tolerant.

- **Spatial analytics:** Spatial data is generated in a multitude of ways. IoT applications have increased such data exponentially. Spatial analytics is the demand of the modern era due to the exponential growth of data that is generated through satellite measurements and imaging, through sensors on IoT applications, use of specialised handheld devices, Wi-Fi, etc. Spatial analytics is the logical processing of such data using topological or geographical properties.

A few popular Spatial analytics open source tools are listed below.

- 1) **GIS tools for Hadoop:** This comprises a collection of GIS tools for Spatial analytics of Big Data. GIS tools for Hadoop are divided into two main parts: Spatial Framework for Hadoop and Geoprocessing Tools for Hadoop. Spatial Framework for Hadoop contains Java Helper utilities for Hadoop developers and Hive spatial user-defined functions. Geoprocessing Tools for Hadoop consists of ArcGIS Desktop or server. ArcGIS supports advanced

analysis, data visualisation and authoritative data maintenance in both 2D and 3D.

- 2) **Geoplot:** This is the high level geospatial data visualisation library for Python. It is an extension of cartopy and matplotlib that makes mapping easy. It has three main features: high-level plotting API, native projection support, and compatibility with matplotlib.

- 3) **Geospark:** This is the cluster computing system for processing large scale spatial data. It extends Apache Spark and SparkSQL with the set of Spatial Resilient Distributed Datasets (SRDDs)/ SpatialSQL.

- 4) **Googletway:** This is the R package for accessing and plotting Google Maps. Googletway provides access to Google Maps APIs, and the ability to plot an interactive Google Map overlaid with various layers and shapes.

- **Time series analytics:** Time series data analytics is familiar in IoT applications such as health monitoring systems, weather forecasting systems, geological applications, and processing manufacturing and other industrial settings.

Several popular Time series analytics open source tools are listed below.

- (1) **Influxdb:** This open source Time series platform includes several APIs for a variety of data analytics tasks such as storing and querying, ETL process, dashboard and visualisations. It is very helpful for real-time analytics.

- (2) **Timescaledb:** This tool is open source and optimised for complex queries in real-time situations. It performs automatic partitioning across time and space. It looks like regular tables but is only an abstraction of many individual tables comprising actual data.

- (3) **Tdengine:** This open source Big Data platform is designed and optimised for the Internet of Things. It provides several functionalities such as caching, stream computing and message queuing.

- (4) **Nightingale:** This open source, distributed high performance system is suitable for IoT applications.

Analytics in IoT is very crucial in the modern era, as billions of connected devices generate a humongous amount of data every day. In this article, we have described three main types of analytics for IoT applications: Streaming, Spatial and Time series. In each category, several popular open source tools that are very essential for effective analytics have been listed. **END** 

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